

MONITORING, EVALUATION AND LEARNING

Inkomoko MEL Endline Outcome Analysis

Sales Growth, Household Savings, Programme
Performance along Theory of Change for refugee
and host-community entrepreneurs.

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837

clients surveyed

610

businesses analysed

+33.6%

growth in mean sales

75%

of households saving



Sales grew by a third and three in four households save; the gaps are about scale and location

+9,011 CFA

mean monthly net sales change

+33.6%

growth on baseline of 26,792 CFA

99%

of 610 businesses sold more

75%

of households save; median 10,000 CFA

Key findings

- Growth is near-universal and about 33% in every subgroup; CFA gaps reflect starting size ($r = 0.70$)
- Location is the widest gap: Sawa +12,505 vs Lala +5,692 CFA
- Gender and refugee status differ little; youth gap is mostly a Lala Camp effect
- Deeper support tiers gain more CFA, but started about twice as large
- Refugees save more (87% vs 66%); Lala savers hold least (9,034 CFA)
- Pathway proven to higher sales and savings; household resilience not yet measured

Priority actions

- Build a Lala Camp growth package: market access, working capital, savings groups
- Evaluate advisory and investment tiers with a comparison group
- Lift business management in Hekima Town
- Add baseline savings, employment and household consumption measures

Every subgroup improved, with the widest differences by location and support

+9,011 CFA

mean net change (95% CI
8,379 to 9,642)

33.6%

overall growth

604 of 610

businesses with higher
sales

75.1%

of households saving

Dimension	Group	n	Baseline sales	Endline sales	Net change	% Growth	% saving	Mean savings
Gender	Female	394	26,488	35,382	+8,894	33.6%	73%	12,379
Gender	Male	216	27,347	36,570	+9,223	33.7%	79%	14,158
Refugee	Refugee	261	26,948	35,831	+8,883	33.0%	87%	12,732
Refugee	Host	349	26,676	35,782	+9,106	34.1%	66%	13,352
Location	Hekima Town	214	26,100	34,661	+8,561	32.8%	63%	13,944
Location	Lala Camp	189	15,598	21,290	+5,692	36.5%	84%	9,034
Location	Sawa Camp	207	37,729	50,235	+12,505	33.1%	80%	16,145
Youth	Youth (35 or under)	287	23,258	31,189	+7,930	34.1%	76%	11,423
Youth	Non-youth	321	29,818	39,816	+9,998	33.5%	75%	14,393
Support	Training	153	19,706	26,096	+6,390	32.4%	76%	11,161
Support	+ advisory	296	22,380	30,081	+7,701	34.4%	75%	13,720
Support	+ investment	161	41,639	55,547	+13,907	33.4%	74%	13,609

Sales in CFA per month; n = businesses with valid baseline and endline sales (youth rows exclude 2 with missing age). Shading: darker = higher.

From 837 surveyed clients to 610 analysed businesses, all built in auditable Excel formulas

837 clients surveyed

186 idea-stage clients: no business yet, so no sales (all Training tier)

651 existing businesses

41 lost: 20 closed and 21 not interviewed. All 41 were Training-only clients

610 analysed

Valid baseline and endline sales; the same 610 answer the savings question

Net change = endline minus baseline monthly sales. **Youth** = age 35 or under. **Savings** = liv_save_hwmuch_CFA, measured at endline only.

How we analysed

- 1 Rebuild**
Clean_Data is formula-driven from untouched Raw_Data, using editable thresholds
- 2 Summarise**
Means, growth, 95% confidence intervals for five dimensions
- 3 Test**
Paired t-test on sales ($t = 27.98$, $p < 0.001$); correlations
- 4 Cross-tabulate**
Tier by location, refugee status by location, gender by location
- 5 Check**
22 data-quality checks and outlier sensitivity

Excel only; formulas include COUNTIFS, AVERAGEIFS, QUARTILE, CORREL, TDIST. Source: Missing_Code_Audit and README sheets.

Cleaning standardised three missing codes and 22 checks confirmed the data is fit for analysis

Missing codes and rules

Code	Cells	Meaning	Rule
999	1,798	Not reached at endline; idea stage	Blank; never averaged
998	2	Age field	Blank; client kept
"N/A"	186	Business type, idea stage	Blank (structural)
Blank	3,391	Skip logic; not reached	No employees = 0; savings among savers

Also fixed

- ❖ 832 ages stored as text, converted to numbers
- ❖ Impossible values blanked: 120,626 employees; a 1 CFA salary
- ❖ All 837 rows kept and flagged; none deleted

12

passed

10

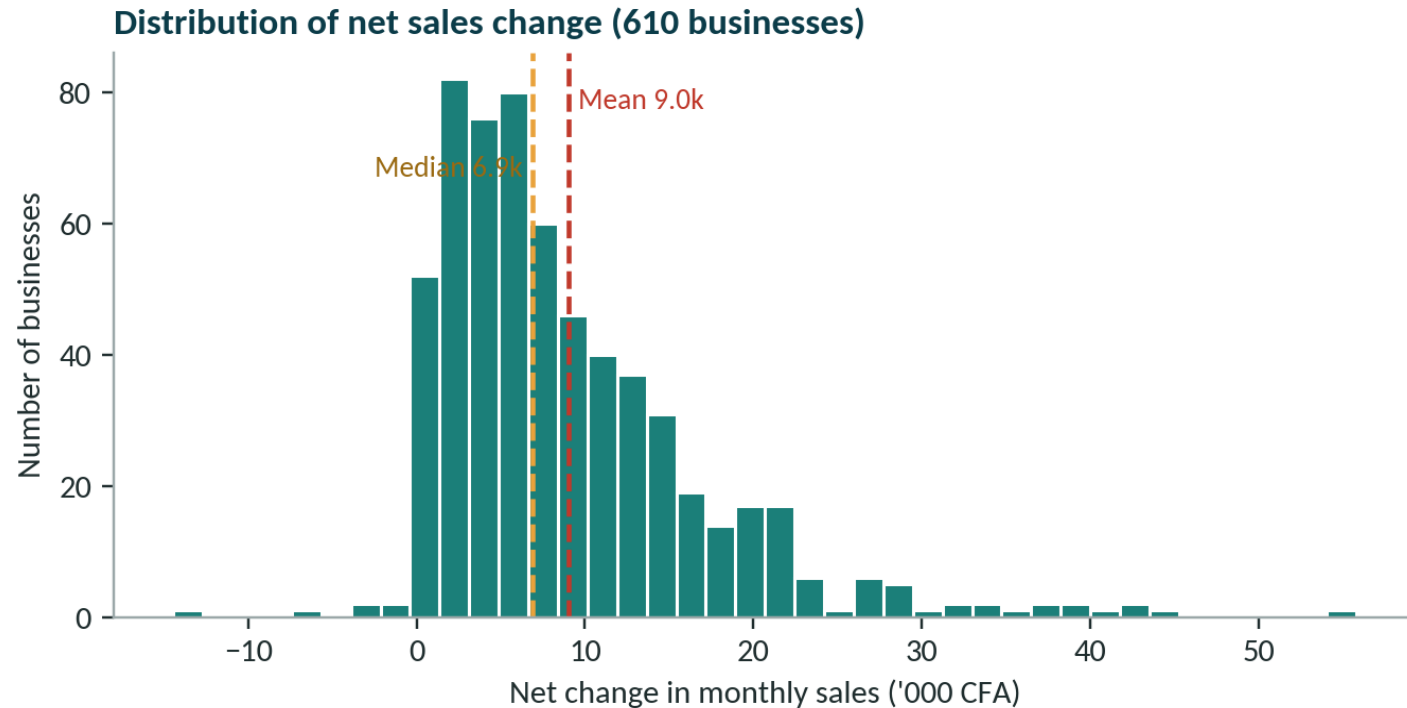
flagged and resolved

What we checked

- ❖ No duplicate IDs or sales pairs
- ❖ Ranges: age, employees, salary, 0 to 10 scale
- ❖ Logic: idea stage has no sales; skip logic holds
- ❖ 5 very large gains and 2 savings above 71,000 CFA flagged and kept
- ❖ Without the 5 gains, mean growth stays 33.0%

Source: Missing_Code_Audit and DQ_Checks sheets. The cleaned dataset is delivered as Inkomoko_Cleaned_Dataset.xlsx.

Monthly sales rose from 26,792 to 35,803 CFA, and 99% of businesses improved



+9,011
mean gain (CFA)

+6,918
median gain (CFA)

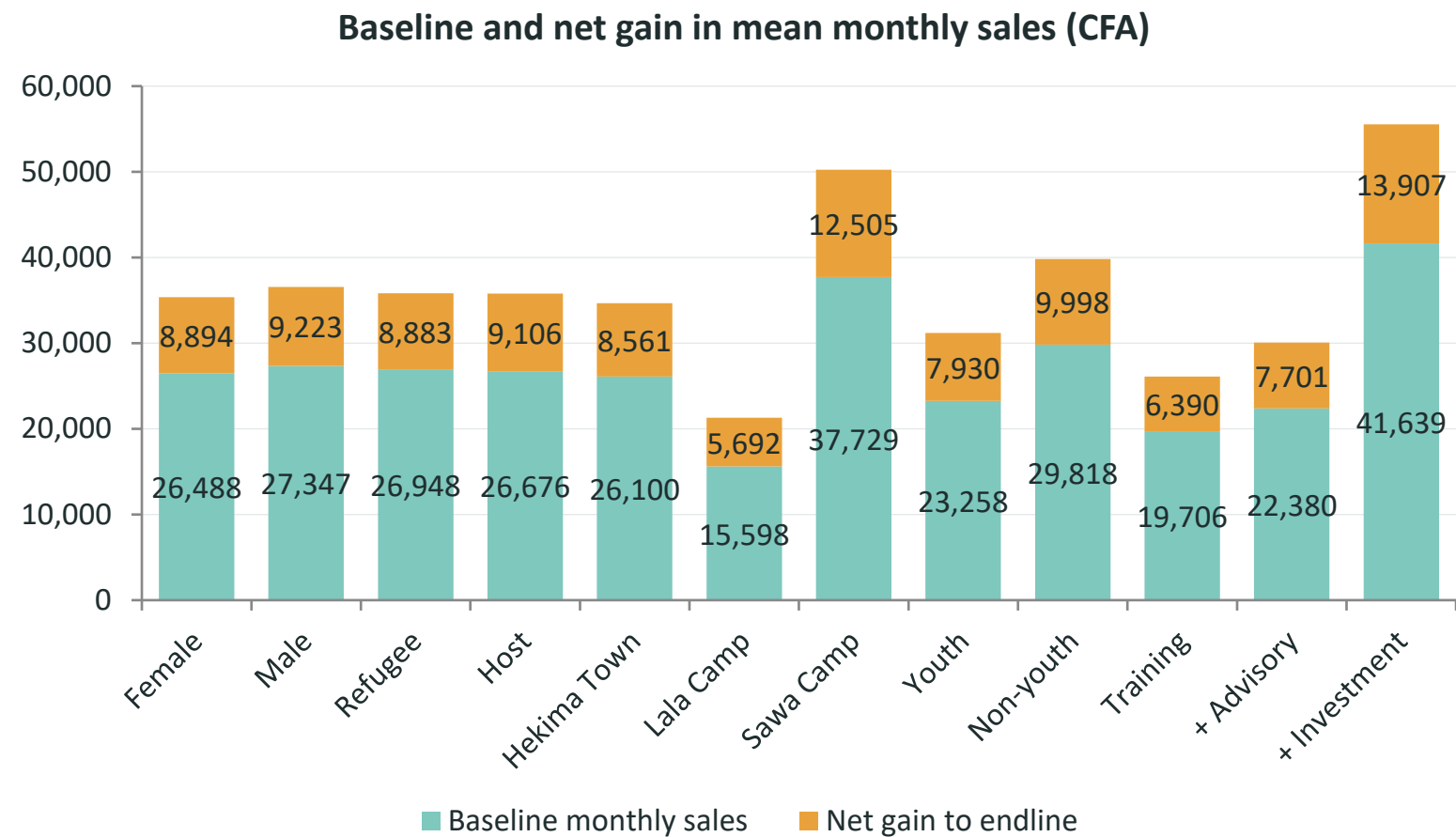
6 of 610
businesses declined

What it shows

- The gain is statistically strong: paired $t = 27.98$, $p < 0.001$
- Right-skewed: the median is 6,918 CFA, below the mean of 9,011
- One in four gained under 3,384 CFA; one in four over 12,578 CFA
- All 6 declines are in Hekima Town (1,500 to 14,433 CFA)
- Before-after design: no comparison group

$n = 610$ existing businesses with valid baseline and endline sales. 95% CI for mean gain: 8,379 to 9,642 CFA. Source: Summary sheet.

Absolute gains are largest in Sawa Camp and the investment tier, which also started largest



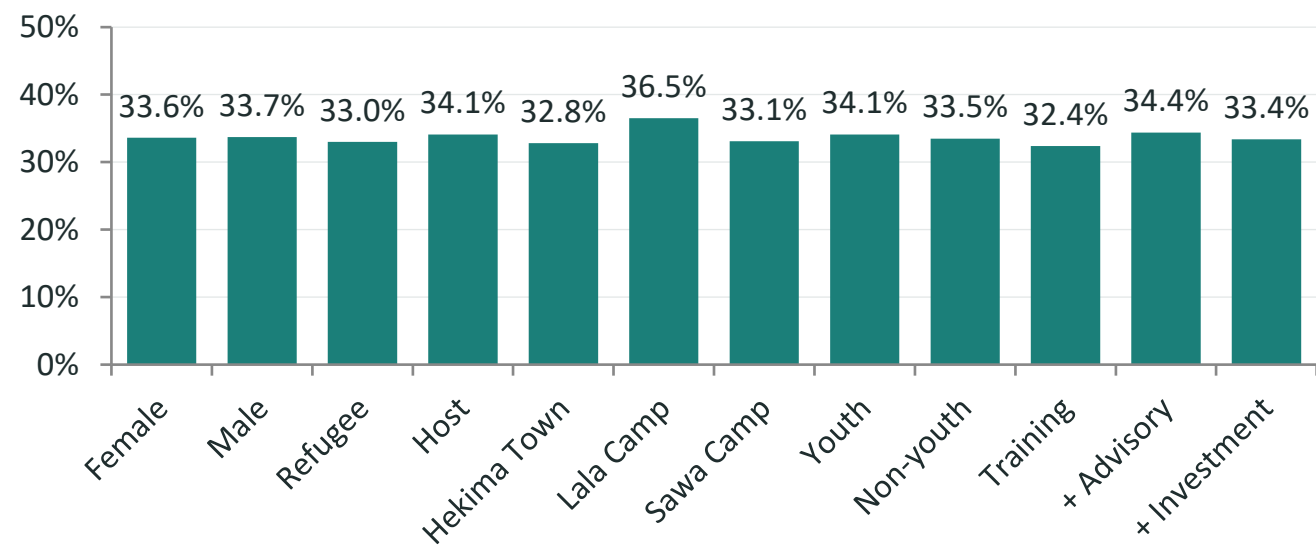
Pattern

- Location: Sawa +12,505 vs Lala +5,692 CFA (2.2x)
- Tier: +6,390, +7,701, +13,907 CFA
- Gender: 8,894 vs 9,223 CFA, overlapping intervals
- Refugee status: 8,883 vs 9,106 CFA
- Youth: 7,930 vs 9,998 CFA, smaller base

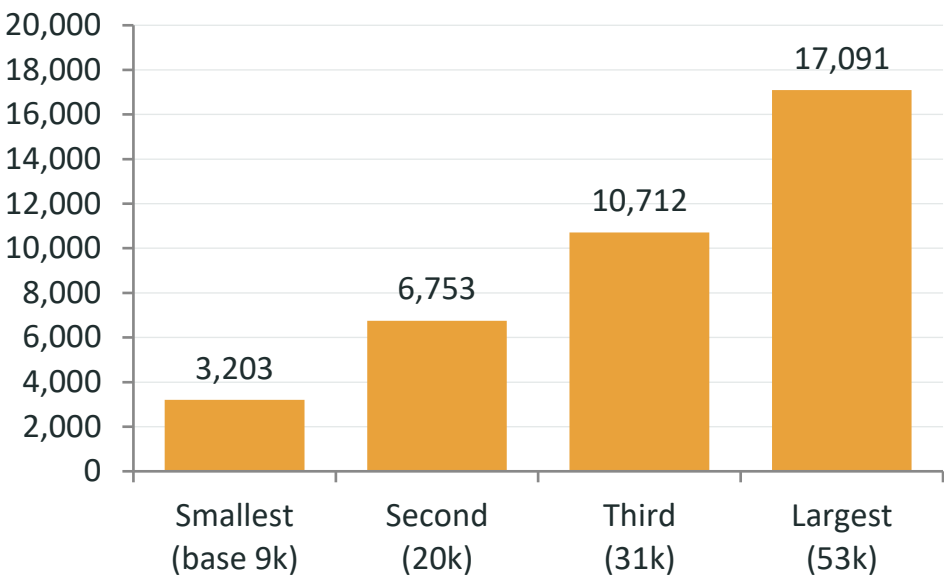
*" + Advisory " = Training & advisory; " + Investment " = Training, advisory, & investment. Top of each bar is the endline mean.
Source: Sales_Analysis sheet.*

Same growth, different scale: businesses grow about a third whatever their starting size

Growth



Mean gain (CFA) by baseline-sales quarter



0.70

correlation of baseline sales with CFA gain

-0.05

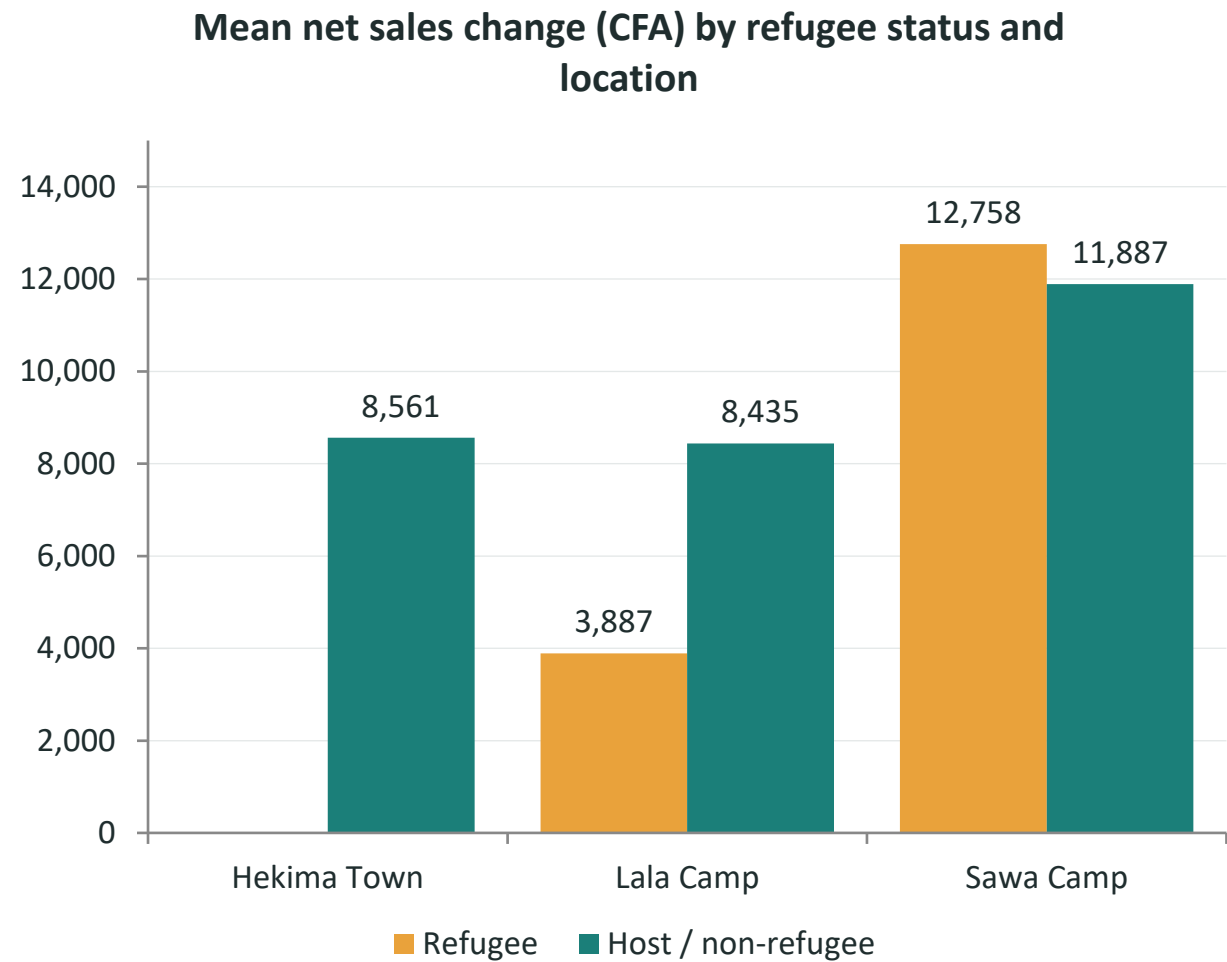
correlation of baseline sales with % growth

31% to 36%

median growth from largest to smallest quarter

*Growth is 32.4% to 36.5% in all 12 subgroups. Lala Camp has the smallest CFA gain but the highest growth (36.5%).
Source: Sales_Analysis sheet and Clean_Data.*

Lala Camp refugees gain least: 3,887 CFA, against 12,758 CFA for refugees in Sawa Camp



Lala Camp refugees

n = 114, baseline 10,788 CFA, gain 3,887 CFA.
Hosts in Lala gain 8,435 CFA.

Access to new funds

Lala Camp 13% and Hekima 19%, against 60% in Sawa Camp.

Hekima Town

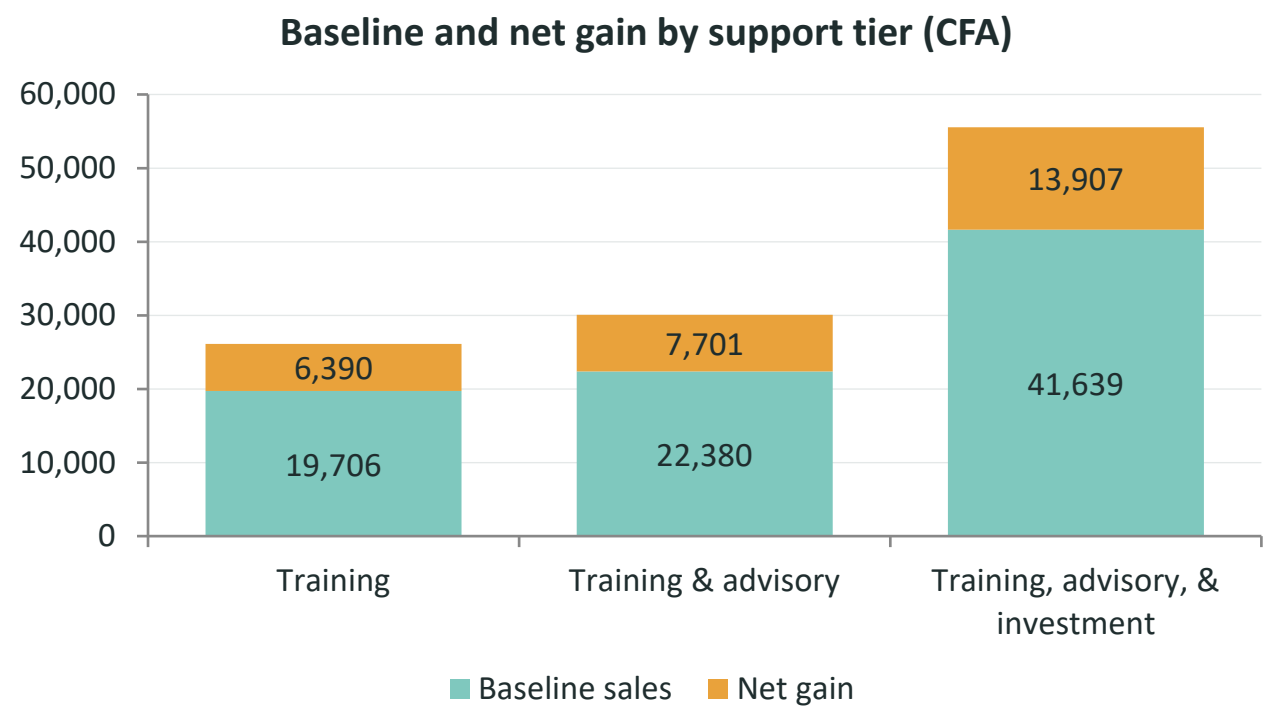
Lowest satisfaction (6.5 of 10), 7% keep records, and all 6 businesses with falling sales

Sawa Camp

Highest baseline (37,729 CFA), best access to funds and the most employers.

*Hekima Town has no refugee clients, so its refugee bar is blank. Sample sizes: Lala refugees 114, hosts 75; Sawa refugees 147, hosts 60.
Source: Crosstabs sheet C.*

Deeper support goes with larger gains, but growth is the same and investment clients started twice as large



Mean net change, CFA (n)

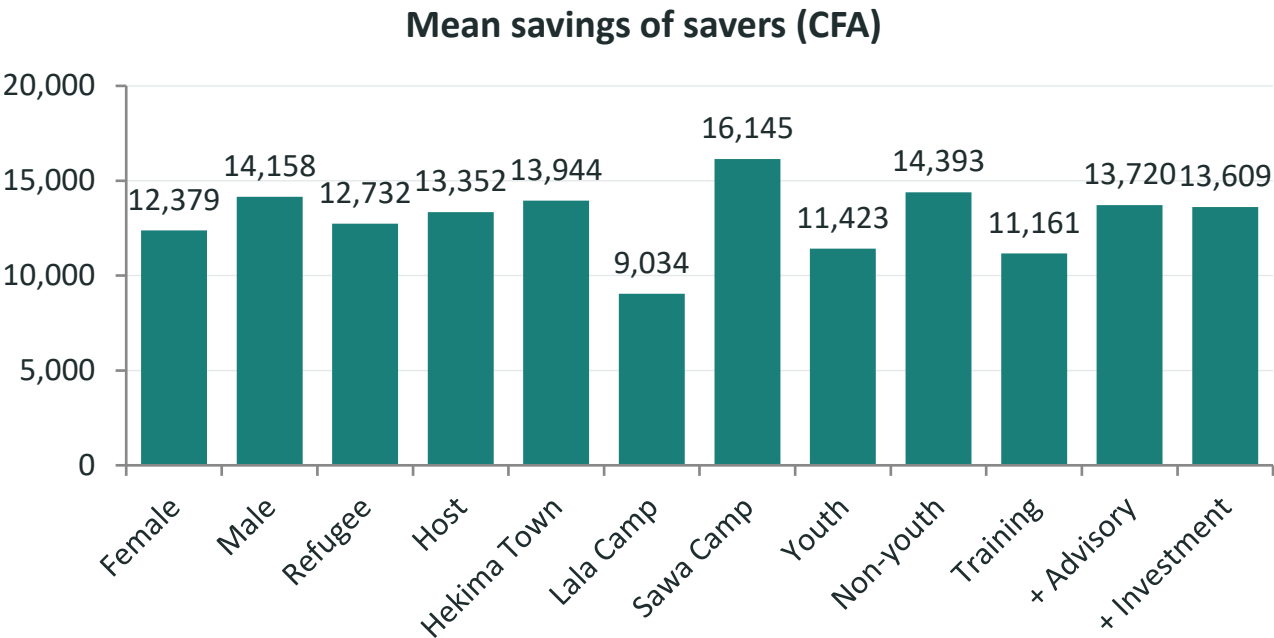
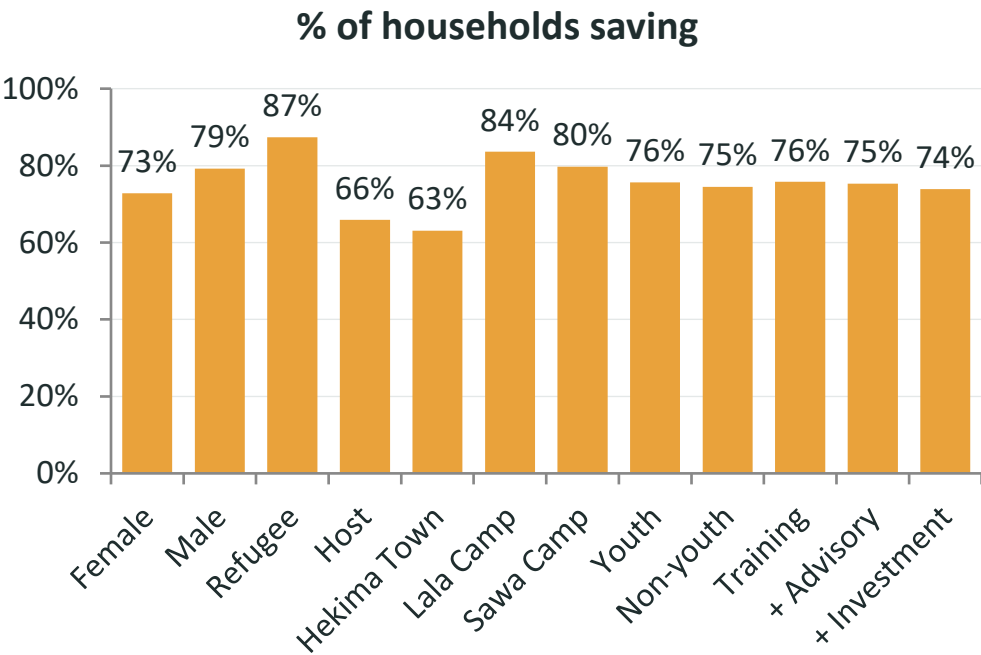
	Hekima	Lala	Sawa
Training	5,933 (55)	3,790 (54)	10,153 (44)
+ Advisory	7,903 (105)	4,150 (83)	10,235 (108)
+ Invest.	12,516 (54)	10,129 (52)	18,846 (55)

- ❖ Investment gains are 1.9x to 2.7x those of Training in every location
- ❖ Advisory alone adds little over training in Lala (4,150 vs 3,790 CFA)

Relative growth is 32.4%, 34.4% and 33.4% across the tiers. Larger CFA gains reflect larger businesses, so the data cannot yet show that advisory or investment raises growth. A phased roll-out or matched comparison would.

Source: Crosstabs sheets A and H, TOC_Indicators sheet.

Three in four households save, refugees most often, but Lala Camp savers hold the least



Who saves

Refugees 87% vs hosts 66%. Lowest: Sawa hosts 48% and Hekima women 58%.

How much

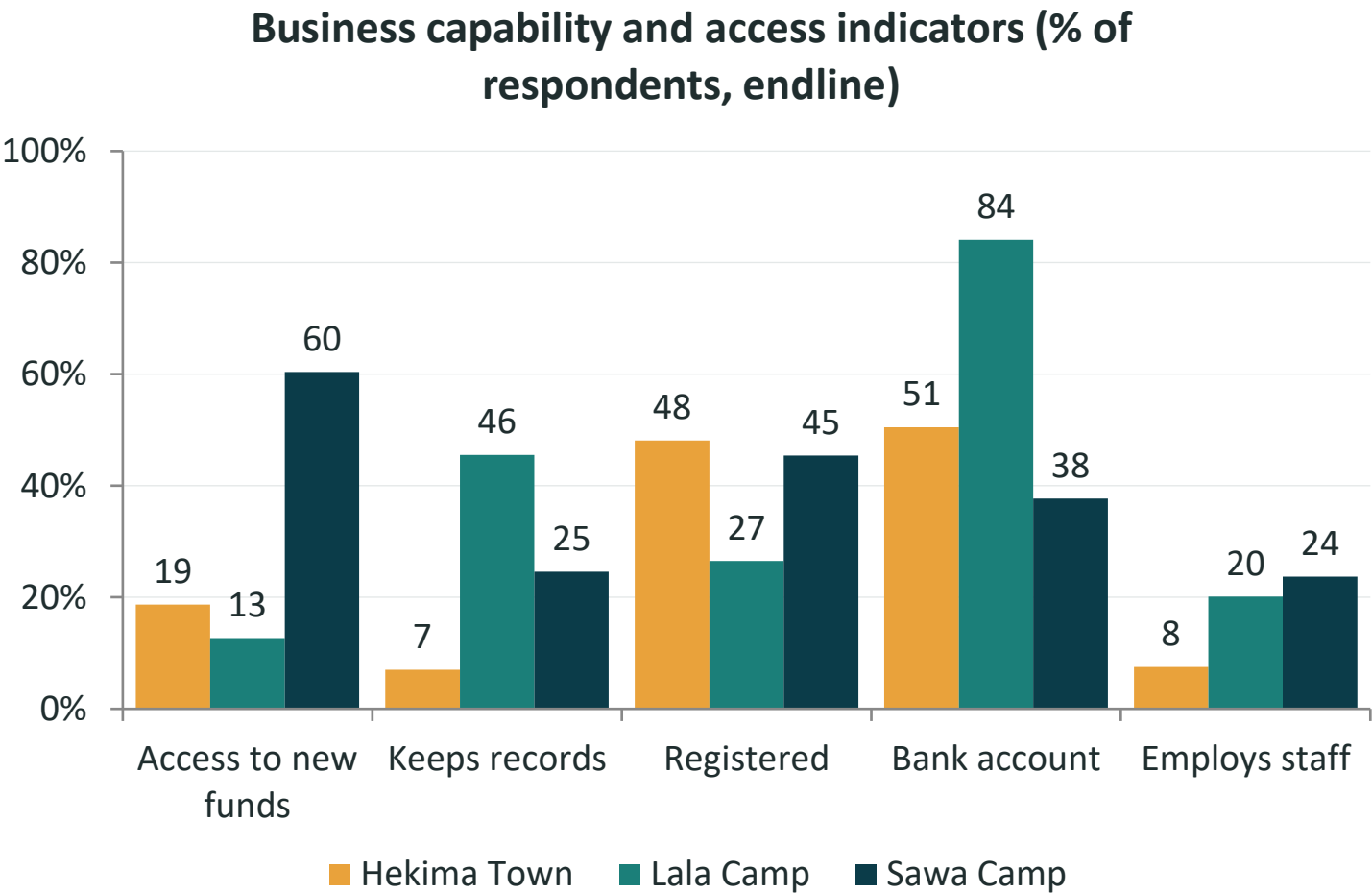
Lala savers 9,034 CFA vs Sawa 16,145 CFA, but 42% of monthly sales against 32%.

What else

Tier barely matters (74% to 76%). Mobile money used by 57% of savers. Savings correlate 0.30 with sales gains.

Savings = liv_save_hwmuch_CFA, measured at endline only, so levels not change. Median saver holds 10,000 CFA. Source: Savings_Analysis sheet.

Each location has a different constraint: Hekima management, Lala finance, Sawa banking



Hekima Town

Only 7% keep records or employ staff; satisfaction 6.5 of 10. Needs business-management support.

Lala Camp

Banked (84%) and records kept (46%), yet only 13% access new funds. Needs capital and markets.

Sawa Camp

Best access to funds (60%) and most employers (24%), lowest bank use (38%). Mobile money and groups fit.

Overall: 17% employ staff, 25% keep records, 57% bank, 41% registered, 31% access new funds. Endline only, no baseline.
Source: Clean_Data sheet.

What is going well, what is not, and what to improve

Going well	Not going well	Improvements needed
✓ 99% of businesses grew, about 34% in every subgroup ✓ Women (65%) and refugees (43%) gain as much as men and hosts ✓ 75% of households save; mobile money leads ✓ Full-package clients most satisfied (7.2 of 10)	✓ Lala refugees gain 3,887 CFA; only 13% access new funds ✓ Hekima: all 6 declines, 7% keep records ✓ Sawa hosts (48%) and Hekima women (58%) save least ✓ 41 lost businesses, all Training-only; only 17% employ staff	✓ Lala package: market access, working capital, savings groups ✓ Hekima: record-keeping and costing in advisory ✓ Savings products for Sawa hosts and Hekima women ✓ Prove the value of advisory and investment tiers

Differences are larger by location and support tier than by gender, refugee status or youth status. Source: Sections 6 to 8 of the analysis.

Six actions turn the findings into programme improvements

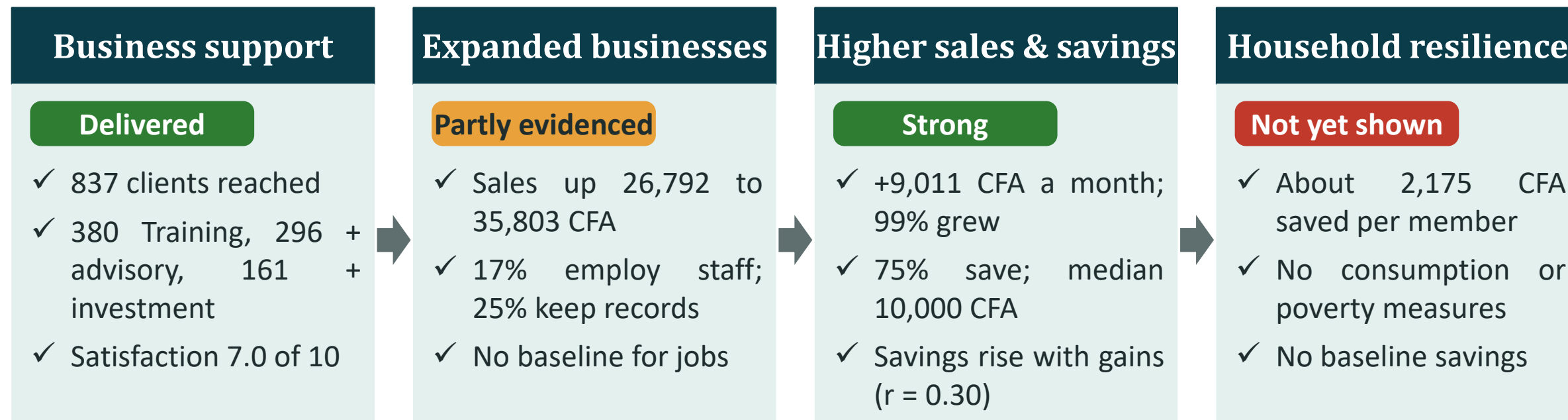
Action	Evidence	Indicator to track
1. Lala Camp growth package	Baseline 15,598 CFA; refugee gain 3,887; 13% access funds	Access to funds; gain for Lala refugees
2. Hekima management uplift	7% keep records; satisfaction 6.5; 6 of 6 declines	Record-keeping %; declining businesses
3. Targeted savings products	Sawa hosts 48% and Hekima women 58% save	% saving in these groups
4. Evaluate advisory and investment	Growth about 33% in all tiers; investment started 2x larger	Growth vs comparison group
5. Reduce attrition	41 of 194 Training clients lost (21%); 186 idea-stage	Follow-up rate by tier
6. Extend measurement	No baseline savings or employment; no consumption data	Resilience indicators covered

Read with care

- ❖ No comparison group: gains may reflect inflation or seasonality
- ❖ Tiers were not randomly assigned
- ❖ All 41 lost businesses were Training-only, which may flatter that tier
- ❖ Savings are measured at endline only
- ❖ Nominal CFA, no inflation adjustment

Leads and timings are for discussion with the programmes team. Source: Sections 9 and 10 of the analysis report.

The theory of change holds up to higher sales and savings; household resilience is not yet shown



Bottom line. Clients are growing and saving across every segment. To complete the pathway, add household consumption and baseline savings, follow clients over time, and use a comparison group so gains can be attributed to Inkomoko.

Pathway status labels are analyst judgement. Employment, records and banking are endline only. Source: TOC_Indicators sheet.

CONCLUSION

- ❑ Strong outcomes.
- ❑ Measured claims.
- ❑ Sharper next steps.

Sales and savings moved decisively; but causality is not proven. Deeper-support effects remain uncertain, while formalization and job creation lag.

NEXT MEASUREMENT PRIORITIES

- ✓ Consumption
- ✓ Assets
- ✓ Poverty
- ✓ Baseline savings
- ✓ Formalisation
- ✓ Stronger follow-up

Source: Inkomoko MEL Endline Outcome Analysis, October 2026

